EE 533 Lab #10
Network Processor Integration and Refinements 2

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NOTE: You must use GitHub to store all source codes and reports associated with the lab assignments. You must commit/push all modifications with detailed descriptions at least once daily while working on the lab assignment. Please submit the GitHub link and a record of your commits, along with detailed commit descriptions, to Coursistant before the due date.

The main purpose of this lab is to refine your core design toward its use in the final project. At the end of the lab, your hardware and firmware system should be functional without many issues. You are expected to have:

(1) Scripts or programs that packetize file(s) containing ANN parameters (weights and biases) and your test data, and then send them to your NetFPGA.
(2) The firmware (CUDA program compiled/translated to run on your SoC) and your hardware (ARM, GPU/Tensor, and other acceleration modules) that processes the packets and then computes the inferences using your scalable hardware accelerators.
(3) As inference outputs are generated, your system should collect and organize data into network packets with correct headers to send out to a destination node.
(4) Scripts or programs that process the output network packets from NetFPGA for analysis and presentations.
(5) You also need to integrate and demonstrate hardware and software mechanisms to precisely evaluate the performance of the system for comparison and for use in determining the potential performance of the architecture when it is scaled up.

While the current design should be optimized to handle bfloat16 weights and biases, you may need to modify it for your final project to implement application-specific improvements.

Please submit the source code, test code, a description of your test setup, and the output results, including screen captures. You must also demonstrate test code running on your network processor on the demo YouTube video.